



Qualification Pack



MEP Executive

Mechanical Systems/ Electrical Systems/ Plumbing Systems

QP Code: ICE/CON/Q1501

Version: 1.0

NSQF Level: 3

Integrated Council for Entrepreneurship and Skilling || 533-R
Model Town Ludhiana || email:cmswami@ice.net.in



Qualification Pack

Contents

ICE/CON/Q1501: MEP Executive	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Elective 1: Mechanical Systems</i>	3
<i>Elective 2: Electrical Systems</i>	3
<i>Elective 3: Plumbing Systems</i>	3
<i>Qualification Pack (QP) Parameters</i>	4
ICE/CON/N1523: Identify Tools, Equipment and Materials Relevant to MEP Work at a Construction Site	6
ICE/CON/N1524: Read and interpret MEP system Drawings and Specifications	12
ICE/CON/N9901: Implement Safe Work Practices and Environmental Stewardship at Construction Sites	19
DGT/VSQ/N0101: Employability Skills (30 Hours)	24
ICE/CON/N1517: Assist in Installation of Mechanical Systems and HVAC Ducts	30
ICE/CON/N1518: Assist in Repair, Maintenance and Testing of Mechanical Systems	35
ICE/CON/N1519: Assist in Electrical Wiring Low Voltage (LV) Installations and Earthing systems	40
ICE/CON/N1520: Assist in Installation, Repair and Maintenance of Electrical Systems	46
ICE/CON/N1521: Assist in Installation of Plumbing Systems	51
ICE/CON/N1522: Assist in Repair and Maintenance of Plumbing Systems	56
Assessment Guidelines and Weightage	60
<i>Assessment Guidelines</i>	60
<i>Assessment Weightage</i>	61
Acronyms	64
Glossary	65



Qualification Pack

ICE/CON/Q1501: MEP Executive

Brief Job Description

The MEP Executive provides essential support in the installation, maintenance and repair of Mechanical, Electrical and Plumbing (MEP) systems. This role involves assisting MEP technicians and engineers with various tasks, including reading blueprints, handling tools, performing basic installations and troubleshooting minor issues in line with sector-specific standards. The individual collaborates with other teams on-site or within facilities to ensure MEP systems operate efficiently, safely and in compliance with industry codes and standards.

Personal Attributes

The individual should be physically fit and mentally alert. The person should have problem-solving, coordination and appropriate verbal and written communication skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ICE/CON/N1523: Identify Tools, Equipment and Materials Relevant to MEP Work at a Construction Site](#)
2. [ICE/CON/N1524: Read and interpret MEP system Drawings and Specifications](#)
3. [ICE/CON/N9901: Implement Safe Work Practices and Environmental Stewardship at Construction Sites](#)
4. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives (mandatory to select at least one):

Elective 1: Mechanical Systems

1. [ICE/CON/N1517: Assist in Installation of Mechanical Systems and HVAC Ducts](#)
2. [ICE/CON/N1518: Assist in Repair, Maintenance and Testing of Mechanical Systems](#)

Elective 2: Electrical Systems

1. [ICE/CON/N1519: Assist in Electrical Wiring Low Voltage \(LV\) Installations and Earthing systems](#)
2. [ICE/CON/N1520: Assist in Installation, Repair and Maintenance of Electrical Systems](#)



Qualification Pack

Elective 3: Plumbing Systems

1. [ICE/CON/N1521: Assist in Installation of Plumbing Systems](#)
2. [ICE/CON/N1522: Assist in Repair and Maintenance of Plumbing Systems](#)

Qualification Pack (QP) Parameters

Sector	Construction
Sub-Sector	Real Estate and Infrastructure Construction
Occupation	MEP (Mechanical, Electrical and Plumbing)
Country	India
NSQF Level	3
Credits	22
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 7412.0202, NCO-2015/7412.0200, NCO-2015/ 7411.0100, NCO-2015/ 7126.0103, NCO-2015/ 7126.0107
Minimum Educational Qualification & Experience	10th grade pass OR 8th grade pass with 2 Years of experience Relevant Industry OR 5th grade pass with 5 Years of experience Relevant Industry OR Previous relevant Qualification of NSQF Level 2 with 1 Year of experience Relevant Industry OR Previous relevant Qualification of NSQF Level 2.5 with 6 Months of experience Relevant Industry
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	Not Applicable
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025



Qualification Pack

Version	1.0
Reference code on NQR	NCVET-QG-03-CO-046422025-V1-ICES
NQR Version	1.0

Remarks:

The Min. Job Entry Age for this Job Role will be as per Government Norms



Qualification Pack

ICE/CON/N1523: Identify Tools, Equipment and Materials Relevant to MEP Work at a Construction Site

Description

This NOS defines the skills and knowledge required to identify, verify and handle tools, equipment and materials used in Mechanical, Electrical and Plumbing (MEP) work at construction sites, ensuring readiness for installation, operation and safe usage under supervision.

Scope

The scope covers the following :

- Identify Tools and Equipment for MEP Task
- Identify Materials Used in MEP Works
- Follow Safety and Storage Procedures for MEP Tools & Materials

Elements and Performance Criteria

Identify Tools and Equipment for MEP Tasks

To be competent, the user/individual on the job must be able to:

- PC1.** identify commonly used hand tools for mechanical works such as spanners, pliers, pipe wrenches, threading tools and cutters.
- PC2.** identify electrical tools including testers, multimeters, screwdrivers, crimping tools, cable cutters, and insulation tools.
- PC3.** identify plumbing tools such as pipe wrenches, PVC/CPVC cutters, threaders, pipe benders, and jointing equipment.
- PC4.** identify power tools relevant to MEP works, including drills, grinders, hydraulic benders, and cable pulling machines.
- PC5.** identify lifting and shifting equipment used for MEP material handling such as chain blocks, trolleys, pulleys and scaffolds.
- PC6.** verify the suitability of tools and equipment as per the assigned task and work method statements.
- PC7.** check tools and equipment for damage, defects or missing components before use.
- PC8.** ensure that tools and equipment are calibrated, labelled and maintained as per manufacturer's guidelines.

Identify Materials Used in MEP Works

To be competent, the user/individual on the job must be able to:

- PC9.** identify commonly used mechanical materials such as ducts, GI sheets, insulation material, fasteners and hangers.
- PC10.** identify electrical materials including cables, conduits, DBS, switches, sockets, MCCB, and earthing components.
- PC11.** identify plumbing materials such as pipes (PVC, CPVC, PPR, GI, HDPE), fittings, valves, sealants, and tapes.
- PC12.** understand the size, type and specification of MEP materials as per drawings and BOQs.



Qualification Pack

- PC13.** differentiate between approved materials and rejected/damaged materials.
- PC14.** follow colour coding, rating and identification markings on electrical and mechanical materials.

Element 3: Follow Safety and Storage Procedures for MEP Tools & Materials

To be competent, the user/individual on the job must be able to:

- PC15.** follow safe storage practices for electrical, mechanical and plumbing materials to avoid damage.
- PC16.** store tools and equipment in designated areas as per site protocols.
- PC17.** use PPE such as gloves, goggles, helmet, reflective jacket and safety shoes while handling MEP materials.
- PC18.** follow correct lifting techniques and manual handling procedures.
- PC19.** report any unsafe conditions, damaged tools, or missing materials to the supervisor.
- PC20.** ensure compliance with site EHS guidelines while identifying or retrieving tools/materials.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company policies for tools and material handling.
- KU2.** standard operating procedures and work method statements for MEP tasks.
- KU3.** storage guidelines for MEP tools and materials.
- KU4.** safety protocols and EHS policies at construction sites.
- KU5.** types of wrenches, spanners, pliers, bolt cutters, threading tools and measurement tools.
- KU6.** power tools such as drills, grinders, benders and cutting machines.
- KU7.** use and interpretation of multimeters, testers, crimping tools, soldering guns, etc.
- KU8.** identification of electrical materials: wires, cables, conduits, DBs, MCBs, earthing strips.
- KU9.** types of pipe cutters, threaders, PPR welding machines and jointing tools.
- KU10.** materials such as pipes, fittings, valves, clamps, sealants and tapes.
- KU11.** colour codes, ratings, gauge and specification readings.
- KU12.** lifting and shifting tools such as trolleys, scaffolds, chain pulleys and hoists.
- KU13.** basic reading of site drawings to identify material specifications.
- KU14.** manufacturer guidelines for tool usage and maintenance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** ability to communicate material and tool requirements to supervisors.
- GS2.** ability to read basic labels, markings and safety notices.
- GS3.** ability to understand sizes, dimensions, gauges and ratings.
- GS4.** ability to compare specifications against drawings/BOQs.
- GS5.** respond to damaged tools/material issues.



Qualification Pack

GS6. identify mismatches in tools/material specifications.

GS7. ability to work safely under minimal supervision.

GS8. follow proper PPE and handling guidelines.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify Tools and Equipment for MEP Tasks</i>	10	20	-	3
PC1. identify commonly used hand tools for mechanical works such as spanners, pliers, pipe wrenches, threading tools and cutters.	-	-	-	-
PC2. identify electrical tools including testers, multimeters, screwdrivers, crimping tools, cable cutters, and insulation tools.	-	-	-	-
PC3. identify plumbing tools such as pipe wrenches, PVC/CPVC cutters, threaders, pipe benders, and jointing equipment.	-	-	-	-
PC4. identify power tools relevant to MEP works, including drills, grinders, hydraulic benders, and cable pulling machines.	-	-	-	-
PC5. identify lifting and shifting equipment used for MEP material handling such as chain blocks, trolleys, pulleys and scaffolds.	-	-	-	-
PC6. verify the suitability of tools and equipment as per the assigned task and work method statements.	-	-	-	-
PC7. check tools and equipment for damage, defects or missing components before use.	-	-	-	-
PC8. ensure that tools and equipment are calibrated, labelled and maintained as per manufacturer's guidelines.	-	-	-	-
<i>Identify Materials Used in MEP Works</i>	10	20	-	4
PC9. identify commonly used mechanical materials such as ducts, GI sheets, insulation material, fasteners and hangers.	-	-	-	-
PC10. identify electrical materials including cables, conduits, DBS, switches, sockets, MCCB, and earthing components.	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. identify plumbing materials such as pipes (PVC, CPVC, PPR, GI, HDPE), fittings, valves, sealants, and tapes.	-	-	-	-
PC12. understand the size, type and specification of MEP materials as per drawings and BOQs.	-	-	-	-
PC13. differentiate between approved materials and rejected/damaged materials.	-	-	-	-
PC14. follow colour coding, rating and identification markings on electrical and mechanical materials.	-	-	-	-
<i>Element 3: Follow Safety and Storage Procedures for MEP Tools & Materials</i>	10	20	-	3
PC15. follow safe storage practices for electrical, mechanical and plumbing materials to avoid damage.	-	-	-	-
PC16. store tools and equipment in designated areas as per site protocols.	-	-	-	-
PC17. use PPE such as gloves, goggles, helmet, reflective jacket and safety shoes while handling MEP materials.	-	-	-	-
PC18. follow correct lifting techniques and manual handling procedures.	-	-	-	-
PC19. report any unsafe conditions, damaged tools, or missing materials to the supervisor.	-	-	-	-
PC20. ensure compliance with site EHS guidelines while identifying or retrieving tools/materials.	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N1523
NOS Name	Identify Tools, Equipment and Materials Relevant to MEP Work at a Construction Site
Sector	Construction
Sub-Sector	
Occupation	MEP (Mechanical, Electrical and Plumbing)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ICE/CON/N1524: Read and interpret MEP system Drawings and Specifications

Description

This NOS covers the ability to read, interpret and apply mechanical, electrical and plumbing (MEP) system drawings, layouts, schematics, legends and technical specifications for installation and site execution work. The job requires strong understanding of MEP symbols, technical standards and interpretation of BOQs, shop drawings, GFC (Good for Construction) drawings and other relevant documentation.

Scope

The scope covers the following :

- Interpret Mechanical (HVAC & Fire-Fighting) Drawings
- Interpret Electrical Drawings & Specifications
- Interpret Plumbing & Public Health Engineering Drawings
- Refer to BOQs, Specifications & Standards
- Apply Drawing Interpretation in Site Execution

Elements and Performance Criteria

Interpret Mechanical (HVAC & Fire-Fighting) Drawings

To be competent, the user/individual on the job must be able to:

- PC1.** identify symbols, line types and notations used in HVAC, fire-fighting and mechanical drawings.
- PC2.** interpret duct routing, pipe routing, equipment layout and mounting locations.
- PC3.** read duct sizes, pipe diameters, insulation details and damper/joint specifications.
- PC4.** interpret mechanical schedules such as airflows, pump heads, equipment capacities and material specifications.
- PC5.** interpret coordination drawings including MEP clearance zones and ceiling space planning.
- PC6.** verify drawing information such as elevations, section IDs and reference levels.

Interpret Electrical Drawings & Specifications

To be competent, the user/individual on the job must be able to:

- PC7.** identify symbols for electrical equipment, lighting, wiring, switches, control systems and earthing.
- PC8.** interpret cable routing, conduit layouts, DB (Distribution Board) schedules and panel diagrams.
- PC9.** read cable sizes, circuit ratings, breaker capacities and load details.
- PC10.** interpret SLD (Single Line Diagram) and schematics for power distribution.
- PC11.** understand electrical layouts such as lighting, power socket, low voltage, CCTV and fire alarm systems.
- PC12.** verify technical compliance with electrical specifications and codes.

Interpret Plumbing & Public Health Engineering Drawings



Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC13.** identify plumbing symbols for water supply, drainage, stormwater and sanitary systems.
- PC14.** interpret pipe routing, pipe sizes, slope, invert levels and fixture locations.
- PC15.** read valve charts, pump schedules, manhole details and plumbing equipment layouts.
- PC16.** understand water tank details, pump room layouts, STP/ETP system drawings and pipe connections.

Refer to BOQs, Specifications & Standards

To be competent, the user/individual on the job must be able to:

- PC17.** interpret BOQ items, material specifications and measurement units.
- PC18.** verify drawing details with project technical specifications.
- PC19.** refer to IS/NBC standards where required.
- PC20.** report discrepancies between drawings and site conditions.

Apply Drawing Interpretation in Site Execution

To be competent, the user/individual on the job must be able to:

- PC21.** mark reference points and routing based on drawing interpretation.
- PC22.** assist in setting out MEP installations as per drawing levels and elevations.
- PC23.** verify that installed equipment, ducts, cables, conduits and pipes match drawing specifications.
- PC24.** maintain updated drawing revisions and follow GFC drawings.
- PC25.** communicate drawing-related issues to supervisors and engineers.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company policies for document control and drawing management.
- KU2.** procedures for obtaining latest revisions of drawings.
- KU3.** work method statements related to MEP installation.
- KU4.** roles and responsibility of design, QA/QC and site teams.
- KU5.** HVAC ducting symbols, pipe symbols, diffuser/grille nomenclature.
- KU6.** firefighting system symbols for sprinklers, hydrants, hose reels, pumps etc.
- KU7.** duct/pipe sizing concepts, insulation thickness, supports and hangers.
- KU8.** mechanical equipment: AHUs, FCUs, VRF, pumps, compressors, cooling towers.
- KU9.** electrical symbols, cable codes, panel diagrams.
- KU10.** SLD, schematic diagrams, distribution board layouts.
- KU11.** understanding of lux levels, load distribution and earthing systems.
- KU12.** PHE system symbols, pipe materials (CPVC, PPR, GI, HDPE).
- KU13.** drainage slope terminology, invert levels, venting systems.
- KU14.** pump room and STP/ETP equipment details.
- KU15.** interpretation of scales, levels, elevations and cross sections.
- KU16.** reading GFC and As-Built drawings.



Qualification Pack

KU17. understanding of reference grids and datum levels.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** ability to communicate material and tool requirements to supervisors.
- GS2.** ability to read basic labels, markings and safety notices.
- GS3.** ability to understand sizes, dimensions, gauges and ratings.
- GS4.** ability to compare specifications against drawings/BOQs.
- GS5.** respond to damaged tools/material issues.
- GS6.** identify mismatches in tools/material specifications.
- GS7.** ability to work safely under minimal supervision.
- GS8.** follow proper PPE and Handling guidelines



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Interpret Mechanical (HVAC & Fire-Fighting) Drawings</i>	6	12	-	2
PC1. identify symbols, line types and notations used in HVAC, fire-fighting and mechanical drawings.	-	-	-	-
PC2. interpret duct routing, pipe routing, equipment layout and mounting locations.	-	-	-	-
PC3. read duct sizes, pipe diameters, insulation details and damper/joint specifications.	-	-	-	-
PC4. interpret mechanical schedules such as airflows, pump heads, equipment capacities and material specifications.	-	-	-	-
PC5. interpret coordination drawings including MEP clearance zones and ceiling space planning.	-	-	-	-
PC6. verify drawing information such as elevations, section IDs and reference levels.	-	-	-	-
<i>Interpret Electrical Drawings & Specifications</i>	6	12	-	2
PC7. identify symbols for electrical equipment, lighting, wiring, switches, control systems and earthing.	-	-	-	-
PC8. interpret cable routing, conduit layouts, DB (Distribution Board) schedules and panel diagrams.	-	-	-	-
PC9. read cable sizes, circuit ratings, breaker capacities and load details.	-	-	-	-
PC10. interpret SLD (Single Line Diagram) and schematics for power distribution.	-	-	-	-
PC11. understand electrical layouts such as lighting, power socket, low voltage, CCTV and fire alarm systems.	-	-	-	-
PC12. verify technical compliance with electrical specifications and codes.	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Interpret Plumbing & Public Health Engineering Drawings</i>	6	12	-	2
PC13. identify plumbing symbols for water supply, drainage, stormwater and sanitary systems.	-	-	-	-
PC14. interpret pipe routing, pipe sizes, slope, invert levels and fixture locations.	-	-	-	-
PC15. read valve charts, pump schedules, manhole details and plumbing equipment layouts.	-	-	-	-
PC16. understand water tank details, pump room layouts, STP/ETP system drawings and pipe connections.	-	-	-	-
<i>Refer to BOQs, Specifications & Standards</i>	6	12	-	2
PC17. interpret BOQ items, material specifications and measurement units.	-	-	-	-
PC18. verify drawing details with project technical specifications.	-	-	-	-
PC19. refer to IS/NBC standards where required.	-	-	-	-
PC20. report discrepancies between drawings and site conditions.	-	-	-	-
<i>Apply Drawing Interpretation in Site Execution</i>	6	12	-	2
PC21. mark reference points and routing based on drawing interpretation.	-	-	-	-
PC22. assist in setting out MEP installations as per drawing levels and elevations.	-	-	-	-
PC23. verify that installed equipment, ducts, cables, conduits and pipes match drawing specifications.	-	-	-	-
PC24. maintain updated drawing revisions and follow GFC drawings.	-	-	-	-
PC25. communicate drawing-related issues to supervisors and engineers.	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N1524
NOS Name	Read and interpret MEP system Drawings and Specifications
Sector	Construction
Sub-Sector	
Occupation	MEP (Mechanical, Electrical and Plumbing)
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ICE/CON/N9901: Implement Safe Work Practices and Environmental Stewardship at Construction Sites

Description

This NOS focuses on enabling construction workers and supervisors to actively adopt safe work practices, promote environmental stewardship and ensure a hazard-free workplace through proactive risk management, sustainable resource usage and timely emergency response.

Scope

The scope covers the following :

- Conduct workplace safety inspections
- Apply safe operational practices
- Promote health, hygiene, and well-being
- Implement sustainability and environmental conservation measures
- Respond effectively to onsite emergencies

Elements and Performance Criteria

Conduct Workplace Safety Inspections

To be competent, the user/individual on the job must be able to:

- PC1.** identify unsafe conditions such as improper scaffolding, poor lighting, open edges or obstructed pathways
- PC2.** assess task-specific risks involving tools, machinery, and hazardous materials
- PC3.** document safety observations and communicate them to the safety lead or supervisor
- PC4.** verify that safety controls such as guard rails, signages and access controls are in place

Apply Safe Operational Practices

To be competent, the user/individual on the job must be able to:

- PC5.** use appropriate PPE and ensure correct fit before beginning any activity
- PC6.** follow safe operation procedures for cutting, lifting, mixing and power tools
- PC7.** implement ergonomic practices to reduce fatigue and prevent musculoskeletal injuries
- PC8.** maintain housekeeping standards by ensuring clean and clutter-free work areas

Promote Health, Hygiene and Well-Being

To be competent, the user/individual on the job must be able to:

- PC9.** follow personal hygiene practices such as handwashing, clean clothing and proper hydration
- PC10.** identify signs of health issues such as heat stress, dehydration and respiratory discomfort
- PC11.** ensure availability and safe usage of sanitation facilities, drinking water and resting zones
- PC12.** report workplace illnesses or physical discomfort to the assigned authority

Implement Sustainability and Environmental Conservation Measures

To be competent, the user/individual on the job must be able to:

- PC13.** follow procedures for safe storage and controlled use of chemicals, fuels and hazardous substances



Qualification Pack

- PC14.** practice material conservation by preventing spillage, wastage and improper handling
- PC15.** undertake waste segregation for recyclable, non-recyclable and hazardous waste
- PC16.** support measures for dust suppression, noise control and energy/water conservation

Respond Effectively to Onsite Emergencies

To be competent, the user/individual on the job must be able to:

- PC17.** identify emergency alarms, muster points and evacuation routes
- PC18.** use basic firefighting equipment such as extinguishers, fire blankets and sand buckets where appropriate
- PC19.** assist in basic first-aid procedures including bleeding control, CPR and heat-stroke management
- PC20.** support emergency communication protocols and follow command instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** workplace safety policies, EHS manuals and site-specific procedures
- KU2.** roles of safety personnel, supervisors and emergency response teams
- KU3.** documentation formats for safety reporting and compliance
- KU4.** internal escalation protocols for hazardous conditions or incidents
- KU5.** legal obligations related to occupational safety and environmental protection
- KU6.** types of workplace hazards chemical, physical, electrical, ergonomic and environmental
- KU7.** safe handling techniques for construction materials, tools and machinery
- KU8.** principles of PPE selection, inspection and maintenance
- KU9.** basics of first aid, CPR and firefighting methods
- KU10.** environmental management practices including waste reduction and pollution control
- KU11.** measures for heat-stress prevention, hydration management and fatigue control
- KU12.** ergonomics and safe manual handling techniques

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret safety signage, manuals and instructions
- GS2.** communicate safety findings and concerns clearly to team members
- GS3.** maintain basic documentation and reports for safety audits
- GS4.** use problem-solving skills to manage minor hazards or disruptions
- GS5.** collaborate effectively during drills, emergency situations and environmental initiatives
- GS6.** demonstrate awareness and discipline in following safe work practices consistently



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct Workplace Safety Inspections</i>	6	12	-	2
PC1. identify unsafe conditions such as improper scaffolding, poor lighting, open edges or obstructed pathways	-	-	-	-
PC2. assess task-specific risks involving tools, machinery, and hazardous materials	-	-	-	-
PC3. document safety observations and communicate them to the safety lead or supervisor	-	-	-	-
PC4. verify that safety controls such as guard rails, signages and access controls are in place	-	-	-	-
<i>Apply Safe Operational Practices</i>	6	12	-	2
PC5. use appropriate PPE and ensure correct fit before beginning any activity	-	-	-	-
PC6. follow safe operation procedures for cutting, lifting, mixing and power tools	-	-	-	-
PC7. implement ergonomic practices to reduce fatigue and prevent musculoskeletal injuries	-	-	-	-
PC8. maintain housekeeping standards by ensuring clean and clutter-free work areas	-	-	-	-
<i>Promote Health, Hygiene and Well-Being</i>	6	12	-	2
PC9. follow personal hygiene practices such as handwashing, clean clothing and proper hydration	-	-	-	-
PC10. identify signs of health issues such as heat stress, dehydration and respiratory discomfort	-	-	-	-
PC11. ensure availability and safe usage of sanitation facilities, drinking water and resting zones	-	-	-	-
PC12. report workplace illnesses or physical discomfort to the assigned authority	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Implement Sustainability and Environmental Conservation Measures</i>	6	12	-	2
PC13. follow procedures for safe storage and controlled use of chemicals, fuels and hazardous substances	-	-	-	-
PC14. practice material conservation by preventing spillage, wastage and improper handling	-	-	-	-
PC15. undertake waste segregation for recyclable, non-recyclable and hazardous waste	-	-	-	-
PC16. support measures for dust suppression, noise control and energy/water conservation	-	-	-	-
<i>Respond Effectively to Onsite Emergencies</i>	6	12	-	2
PC17. identify emergency alarms, muster points and evacuation routes	-	-	-	-
PC18. use basic firefighting equipment such as extinguishers, fire blankets and sand buckets where appropriate	-	-	-	-
PC19. assist in basic first-aid procedures including bleeding control, CPR and heat-stroke management	-	-	-	-
PC20. support emergency communication protocols and follow command instructions	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N9901
NOS Name	Implement Safe Work Practices and Environmental Stewardship at Construction Sites
Sector	Construction
Sub-Sector	
Occupation	Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team



Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services



Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/09/2026
NSQC Clearance Date	12/03/2026



Qualification Pack

ICE/CON/N1517: Assist in Installation of Mechanical Systems and HVAC Ducts

Description

This unit specifies the competencies required to assist in installing mechanical systems such as HVAC units, ducts, pumps, and motors. It focuses on laying out, aligning, and securing mechanical components under supervision.

Scope

The scope covers the following :

- Assist in installing mechanical systems
- Assist with HVAC duct and ventilation installations

Elements and Performance Criteria

Assist in installation of mechanical systems

To be competent, the user/individual on the job must be able to:

- PC1.** identify and arrange required tools, equipment and materials at the construction site
- PC2.** assist in assembling HVAC units as per design specifications
- PC3.** support the alignment and installation of HVAC units, ensuring they are level and securely mounted
- PC4.** lay out pipework and duct connections as per supervisor's instructions
- PC5.** handle basic fittings such as fasteners, brackets and connectors
- PC6.** carry out proper sealing of joints and insulation
- PC7.** follow manufacturer guidelines and supervisor's instructions during the installation
- PC8.** report any discrepancies or missing components to the supervisor

Assist in installation of HVAC duct and ventilation systems

To be competent, the user/individual on the job must be able to:

- PC9.** measure and mark locations for duct installations based on blueprints or layouts
- PC10.** assist in cutting and assembling duct sections using the relevant tools like snips and crimpers
- PC11.** secure ducts to ceilings or walls using hangers or brackets
- PC12.** ensure proper alignment and sealing of duct connections
- PC13.** apply insulation to ducts where required to minimize energy loss
- PC14.** check for obstructions in the airflow and report findings to the supervisor

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** basics of mechanical systems and components



Qualification Pack

- KU2.** tools and equipment used in mechanical system installations
- KU3.** terminologies related to HVAC and ducts
- KU4.** standard safety practices for handling mechanical systems
- KU5.** importance of sealing and insulating HVAC ducts
- KU6.** duct layouts and connection methods
- KU7.** process of marking and measuring for installations
- KU8.** methods for the disposal of waste materials generated during installations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively with team members and supervisors
- GS2.** read and interpret technical documents and blueprints
- GS3.** maintain accurate documentation of installation tasks
- GS4.** solve minor issues related to alignment or fittings
- GS5.** manage time effectively to complete tasks efficiently
- GS6.** follow workplace safety protocols consistently



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Assist in installation of mechanical systems</i>	15	30	-	5
PC1. identify and arrange required tools, equipment and materials at the construction site	-	-	-	-
PC2. assist in assembling HVAC units as per design specifications	-	-	-	-
PC3. support the alignment and installation of HVAC units, ensuring they are level and securely mounted	-	-	-	-
PC4. lay out pipework and duct connections as per supervisor's instructions	-	-	-	-
PC5. handle basic fittings such as fasteners, brackets and connectors	-	-	-	-
PC6. carry out proper sealing of joints and insulation	-	-	-	-
PC7. follow manufacturer guidelines and supervisor's instructions during the installation	-	-	-	-
PC8. report any discrepancies or missing components to the supervisor	-	-	-	-
<i>Assist in installation of HVAC duct and ventilation systems</i>	15	30	-	5
PC9. measure and mark locations for duct installations based on blueprints or layouts	-	-	-	-
PC10. assist in cutting and assembling duct sections using the relevant tools like snips and crimpers	-	-	-	-
PC11. secure ducts to ceilings or walls using hangers or brackets	-	-	-	-
PC12. ensure proper alignment and sealing of duct connections	-	-	-	-
PC13. apply insulation to ducts where required to minimize energy loss	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. check for obstructions in the airflow and report findings to the supervisor	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N1517
NOS Name	Assist in Installation of Mechanical Systems and HVAC Ducts
Sector	Construction
Sub-Sector	
Occupation	MEP (Mechanical, Electrical and Plumbing)
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ICE/CON/N1518: Assist in Repair, Maintenance and Testing of Mechanical Systems

Description

This unit specifies the competencies required to conduct basic repair and maintenance of mechanical systems and assist with system start-up and shutdown processes under supervision.

Scope

The scope covers the following :

- Conduct basic mechanical maintenance
- Assist in installing pump and motor
- Assist in mechanical system start-up and shutdown

Elements and Performance Criteria

Conduct basic mechanical maintenance

To be competent, the user/individual on the job must be able to:

- PC1.** perform visual inspections of mechanical systems for leaks, damage, or wear
- PC2.** clean components such as filters, grilles, and duct surfaces
- PC3.** replace basic components such as air filters, belts, and fasteners as instructed
- PC4.** carry out lubrication of mechanical parts as per operational needs
- PC5.** report unusual noises or system irregularities to the supervisor
- PC6.** assist in maintaining the logs of maintenance activities

Assist in repair and maintenance of pump and motor

To be competent, the user/individual on the job must be able to:

- PC7.** identify the construction site and prepare it for repair and maintenance
- PC8.** assist in the positioning and alignment of pumps and motors using appropriate tools
- PC9.** secure pumps and motors to their foundations or mounts
- PC10.** connect basic electrical or mechanical couplings as instructed
- PC11.** check for leaks or misalignments after installation and report them to the technician

Assist in mechanical system start-up and shutdown

To be competent, the user/individual on the job must be able to:

- PC12.** assist in verifying system connections before start-up
- PC13.** follow the applicable safety protocols during the start-up process
- PC14.** record system parameters such as pressure, temperature and flow rates
- PC15.** report anomalies such as irregular noise, vibration or system errors
- PC16.** assist in shutdown procedures by following step-by-step instructions
- PC17.** clean and store tools and materials after tasks
- PC18.** assist in preparing handover documentation or reports



Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** basics of mechanical systems maintenance
- KU2.** components of pumps and motors (e.g., impellers, bearings)
- KU3.** methods of alignment and mounting pumps and motors
- KU4.** importance of lubrication and cleaning in maintenance
- KU5.** standard procedures for system start-up and shutdown
- KU6.** parameters to monitor during system operations (e.g. pressure, flow)
- KU7.** techniques for identifying system anomalies like noise or vibration
- KU8.** safety measures for handling pumps, motors, and other mechanical components

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively to report maintenance issues and anomalies
- GS2.** maintain clear and accurate maintenance logs
- GS3.** coordinate with technicians to ensure task efficiency
- GS4.** identify and address minor problems during maintenance
- GS5.** adhere to safety guidelines during start-up and shutdown processes



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct basic mechanical maintenance</i>	10	20	-	3
PC1. perform visual inspections of mechanical systems for leaks, damage, or wear	-	-	-	-
PC2. clean components such as filters, grilles, and duct surfaces	-	-	-	-
PC3. replace basic components such as air filters, belts, and fasteners as instructed	-	-	-	-
PC4. carry out lubrication of mechanical parts as per operational needs	-	-	-	-
PC5. report unusual noises or system irregularities to the supervisor	-	-	-	-
PC6. assist in maintaining the logs of maintenance activities	-	-	-	-
<i>Assist in repair and maintenance of pump and motor</i>	10	20	-	4
PC7. identify the construction site and prepare it for repair and maintenance	-	-	-	-
PC8. assist in the positioning and alignment of pumps and motors using appropriate tools	-	-	-	-
PC9. secure pumps and motors to their foundations or mounts	-	-	-	-
PC10. connect basic electrical or mechanical couplings as instructed	-	-	-	-
PC11. check for leaks or misalignments after installation and report them to the technician	-	-	-	-
<i>Assist in mechanical system start-up and shutdown</i>	10	20	-	3
PC12. assist in verifying system connections before start-up	-	-	-	-
PC13. follow the applicable safety protocols during the start-up process	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. record system parameters such as pressure, temperature and flow rates	-	-	-	-
PC15. report anomalies such as irregular noise, vibration or system errors	-	-	-	-
PC16. assist in shutdown procedures by following step-by-step instructions	-	-	-	-
PC17. clean and store tools and materials after tasks	-	-	-	-
PC18. assist in preparing handover documentation or reports	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N1518
NOS Name	Assist in Repair, Maintenance and Testing of Mechanical Systems
Sector	Construction
Sub-Sector	
Occupation	MEP (Mechanical, Electrical and Plumbing)
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ICE/CON/N1519: Assist in Electrical Wiring Low Voltage (LV) Installations and Earthing systems

Description

This unit is about assisting in electrical wiring and installations at construction sites, focusing on low-voltage systems, basic wiring tasks and following instructions under supervision.

Scope

The scope covers the following :

- Prepare for electrical works
- Assist in wiring tasks
- Assist in LV installations
- Assist in testing
- Carry out post-installation tasks

Elements and Performance Criteria

Prepare the site for electrical system installation

To be competent, the user/individual on the job must be able to:

- PC1.** interpret basic site layout plans and identify wiring requirements under supervision
- PC2.** identify and organize tools, equipment, materials and PPE required for electrical wiring tasks
- PC3.** check the quality of wires, conduits, connectors and other components before use
- PC4.** clean and prepare work areas for safe wiring operations

Assist in wiring tasks

To be competent, the user/individual on the job must be able to:

- PC5.** assist in laying conduits and routing cables as per supervisor instructions
- PC6.** pass electrical wires through conduits, ensuring proper alignment and secure placement
- PC7.** position cables, trays and fixtures during installation as directed
- PC8.** assist in fixing wiring to junction boxes, panels and outlets using tools
- PC9.** follow standard color codes and markings for wire identification
- PC10.** support in securing and bundling cables for safety and organization
- PC11.** assist in basic wiring connections for lighting, appliances and switches

Assist in installations of LV system

To be competent, the user/individual on the job must be able to:

- PC12.** assist in the installation of low-voltage electrical circuits for lighting systems
- PC13.** assist in carrying out wiring for HVAC systems under supervision
- PC14.** connect wiring for standby generators following the safety guidelines and supervisor's instructions

Assist in Earthing System Installation

To be competent, the user/individual on the job must be able to:



Qualification Pack

- PC15.** assist in digging pits and laying earthing electrodes as per site specifications.
- PC16.** support in connecting earthing conductors to grounding rods, plates, or mesh.
- PC17.** ensure proper placement of earthing components to achieve minimum resistance.
- PC18.** assist in checking earthing continuity and effectiveness under supervision.

Assist in testing

To be competent, the user/individual on the job must be able to:

- PC19.** conduct basic insulation checks under supervision using tools like a multimeter
- PC20.** follow the applicable safety procedures while handling live wires and connecting wires to electrical panels
- PC21.** report any damaged wires, tools or components to the supervisor immediately

Carry out post-installation tasks

To be competent, the user/individual on the job must be able to:

- PC22.** assist in labelling and documenting wiring connections as required
- PC23.** clean the site by disposing waste materials in designated areas
- PC24.** store unused wiring and equipment in the proper storage areas

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** types of electrical systems commonly used at construction sites
- KU2.** components of electrical systems such as wires, conduits, junction boxes etc.
- KU3.** basics of low-voltage electrical wiring systems
- KU4.** wiring standards, including colour-coding and labelling conventions
- KU5.** common tools used for electrical wiring, e.g. wire strippers, cutters, crimping tools, etc.
- KU6.** handling and maintenance of tools and equipment
- KU7.** types and properties of wires, conduits, and fittings
- KU8.** use of relevant PPE for electrical tasks
- KU9.** hazards associated with electrical wiring and methods to mitigate risks
- KU10.** importance of grounding and earthing in electrical installations
- KU11.** principles of earthing, including types (plate earthing, rod earthing etc.).
- KU12.** components of an earthing system and their functions.
- KU13.** basic steps for assisting in electrical installations
- KU14.** techniques for organizing and routing cables efficiently
- KU15.** reporting procedures for damaged equipment or hazardous conditions

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** convey and interpret technical information, instructions and feedback with team members and supervisors



Qualification Pack

- GS2.** analyze work requirements, assess material suitability and choose the best tools, methods and techniques to meet project goals
- GS3.** record measurements, material usage and quality checks systematically to ensure accuracy and traceability
- GS4.** identify and address challenges, such as misaligned components or damaged materials, through appropriate solutions
- GS5.** collaborate with peers and supervisors to ensure smooth workflow and alignment with project objectives
- GS6.** prioritize tasks and efficiently manage resources to meet deadlines without compromising quality
- GS7.** maintain accuracy in measurements, alignment and finishing to meet design specifications and quality standards
- GS8.** adjust to changing requirements or unforeseen challenges, ensuring project continuity and compliance
- GS9.** foster positive working relationships and facilitate cooperation among team members for improved outcomes
- GS10.** promote and adhere to safety standards, ensuring a secure working environment for oneself and others



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare the site for electrical system installation</i>	5	10	-	2
PC1. interpret basic site layout plans and identify wiring requirements under supervision	-	-	-	-
PC2. identify and organize tools, equipment, materials and PPE required for electrical wiring tasks	-	-	-	-
PC3. check the quality of wires, conduits, connectors and other components before use	-	-	-	-
PC4. clean and prepare work areas for safe wiring operations	-	-	-	-
<i>Assist in wiring tasks</i>	5	10	-	2
PC5. assist in laying conduits and routing cables as per supervisor instructions	-	-	-	-
PC6. pass electrical wires through conduits, ensuring proper alignment and secure placement	-	-	-	-
PC7. position cables, trays and fixtures during installation as directed	-	-	-	-
PC8. assist in fixing wiring to junction boxes, panels and outlets using tools	-	-	-	-
PC9. follow standard color codes and markings for wire identification	-	-	-	-
PC10. support in securing and bundling cables for safety and organization	-	-	-	-
PC11. assist in basic wiring connections for lighting, appliances and switches	-	-	-	-
<i>Assist in installations of LV system</i>	5	10	-	2
PC12. assist in the installation of low-voltage electrical circuits for lighting systems	-	-	-	-
PC13. assist in carrying out wiring for HVAC systems under supervision	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. connect wiring for standby generators following the safety guidelines and supervisor's instructions	-	-	-	-
<i>Assist in Earthing System Installation</i>	5	10	-	2
PC15. assist in digging pits and laying earthing electrodes as per site specifications.	-	-	-	-
PC16. support in connecting earthing conductors to grounding rods, plates, or mesh.	-	-	-	-
PC17. ensure proper placement of earthing components to achieve minimum resistance.	-	-	-	-
PC18. assist in checking earthing continuity and effectiveness under supervision.	-	-	-	-
<i>Assist in testing</i>	5	10	-	1
PC19. conduct basic insulation checks under supervision using tools like a multimeter	-	-	-	-
PC20. follow the applicable safety procedures while handling live wires and connecting wires to electrical panels	-	-	-	-
PC21. report any damaged wires, tools or components to the supervisor immediately	-	-	-	-
<i>Carry out post-installation tasks</i>	5	10	-	1
PC22. assist in labelling and documenting wiring connections as required	-	-	-	-
PC23. clean the site by disposing waste materials in designated areas	-	-	-	-
PC24. store unused wiring and equipment in the proper storage areas	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N1519
NOS Name	Assist in Electrical Wiring Low Voltage (LV) Installations and Earthing systems
Sector	Construction
Sub-Sector	
Occupation	MEP (Mechanical, Electrical and Plumbing)
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ICE/CON/N1520: Assist in Installation, Repair and Maintenance of Electrical Systems

Description

This unit focuses on assisting in the installation and maintenance of temporary electrical setups at construction sites, including lighting systems and low-voltage electrical panels.

Scope

The scope covers the following :

- Prepare for temporary installations
- Assist in installations
- Maintain the temporary electrical systems
- Carry out post-installation and maintenance activities

Elements and Performance Criteria

Prepare for temporary installations

To be competent, the user/individual on the job must be able to:

- PC1.** assess site requirements for temporary electrical setups under supervisor guidance
- PC2.** identify tools and equipment needed for temporary lighting and panel installations
- PC3.** inspect lighting fixtures, cables and panels for functionality before installation

Assist in installation of electrical systems

To be competent, the user/individual on the job must be able to:

- PC4.** assist in laying cables for temporary lighting and power arrangements
- PC5.** position and hold fixtures, lights and panels during installation
- PC6.** assist in connecting temporary electrical panels to power sources like generators
- PC7.** route and secure cables to avoid tripping hazards or damage
- PC8.** assist in installing temporary lighting systems for pathways, work areas and emergency lighting
- PC9.** ensure proper grounding of temporary setups as instructed
- PC10.** follow supervisor instructions to balance power loads on temporary electrical panels

Perform repair and maintenance of the temporary electrical systems

To be competent, the user/individual on the job must be able to:

- PC11.** perform visual inspections of temporary lighting and power setups for damage
- PC12.** replace bulbs, switches, and sockets as instructed
- PC13.** check cables and connectors for wear and tear and report issues to the supervisor
- PC14.** assist in performing basic electrical tests like voltage and current check
- PC15.** isolate circuits safely before starting any maintenance tasks

Carry out post-installation and maintenance activities

To be competent, the user/individual on the job must be able to:



Qualification Pack

- PC16.** assist in disassembling temporary electrical setups once the work is completed
- PC17.** store cables, tools and components in the designated storage areas after use
- PC18.** document maintenance tasks and reported issues for reference

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The purpose and components of temporary lighting systems
- KU2.** Their use of low-voltage electrical panels in construction setups
- KU3.** The importance of grounding and load balancing in temporary setups
- KU4.** The tools used for installing and maintaining temporary systems
- KU5.** The common issues in temporary setups and their troubleshooting methods
- KU6.** The techniques for testing electrical systems using basic tools
- KU7.** The hazards associated with temporary electrical setups and the appropriate safety measures
- KU8.** The importance of isolating circuits before maintenance
- KU9.** The protocols for handling live circuits safely
- KU10.** The processes of temporary lighting and panel installation
- KU11.** The procedures for maintaining and repairing temporary electrical setups

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively with team members and supervisors to understand and execute tasks
- GS2.** read and interpret technical documents, drawings, and schedules accurately
- GS3.** perform basic mathematical calculations required for measurements and material estimation
- GS4.** use the digital tools and software relevant to metal fabrication and reinforcement work
- GS5.** identify and resolve common operational issues during fabrication and installation processes
- GS6.** follow organizational safety protocols and use personal protective equipment (PPE) appropriately
- GS7.** plan and organize work efficiently to meet project timelines and quality standards
- GS8.** maintain tools, equipment and workspaces in a clean and operational condition
- GS9.** adapt to new technologies and methods in construction and fabrication
- GS10.** record and report work progress, materials used and any deviations to supervisors



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for temporary installations</i>	8	15	-	3
PC1. assess site requirements for temporary electrical setups under supervisor guidance	-	-	-	-
PC2. identify tools and equipment needed for temporary lighting and panel installations	-	-	-	-
PC3. inspect lighting fixtures, cables and panels for functionality before installation	-	-	-	-
<i>Assist in installation of electrical systems</i>	8	15	-	3
PC4. assist in laying cables for temporary lighting and power arrangements	-	-	-	-
PC5. position and hold fixtures, lights and panels during installation	-	-	-	-
PC6. assist in connecting temporary electrical panels to power sources like generators	-	-	-	-
PC7. route and secure cables to avoid tripping hazards or damage	-	-	-	-
PC8. assist in installing temporary lighting systems for pathways, work areas and emergency lighting	-	-	-	-
PC9. ensure proper grounding of temporary setups as instructed	-	-	-	-
PC10. follow supervisor instructions to balance power loads on temporary electrical panels	-	-	-	-
<i>Perform repair and maintenance of the temporary electrical systems</i>	7	15	-	2
PC11. perform visual inspections of temporary lighting and power setups for damage	-	-	-	-
PC12. replace bulbs, switches, and sockets as instructed	-	-	-	-
PC13. check cables and connectors for wear and tear and report issues to the supervisor	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. assist in performing basic electrical tests like voltage and current check	-	-	-	-
PC15. isolate circuits safely before starting any maintenance tasks	-	-	-	-
<i>Carry out post-installation and maintenance activities</i>	7	15	-	2
PC16. assist in disassembling temporary electrical setups once the work is completed	-	-	-	-
PC17. store cables, tools and components in the designated storage areas after use	-	-	-	-
PC18. document maintenance tasks and reported issues for reference	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N1520
NOS Name	Assist in Installation, Repair and Maintenance of Electrical Systems
Sector	Construction
Sub-Sector	
Occupation	MEP (Mechanical, Electrical and Plumbing)
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ICE/CON/N1521: Assist in Installation of Plumbing Systems

Description

This unit covers the competencies and knowledge requirements concerning the installation of plumbing systems, including fixtures, pipes, and IoT-based water management systems, under supervision.

Scope

The scope covers the following :

- Prepare for plumbing installations
- Assist in installing plumbing systems
- Assist in installing IoT-based water management systems

Elements and Performance Criteria

Prepare for plumbing installations

To be competent, the user/individual on the job must be able to:

- PC1.** unpack and inspect pipes, fittings and fixtures as per safety and manufacturer guidelines
- PC2.** check the availability and specifications of materials, tools and equipment
- PC3.** identify missing or defective items and report them promptly
- PC4.** clean and organize the work area, ensuring tools and equipment are positioned for ease of access

Assist in installation of plumbing systems

To be competent, the user/individual on the job must be able to:

- PC5.** use basic hand and power tools for cutting, bending and shaping pipes
- PC6.** assist in measuring and marking installation points
- PC7.** hold and position pipes and fixtures during installation
- PC8.** apply basic jointing techniques, such as threading, gluing and sealing under supervision
- PC9.** secure installed fixtures and pipes using clamps and fasteners
- PC10.** assist in testing plumbing systems for leaks and other issues as instructed
- PC11.** ensure proper insulation and alignment of pipes during installation

Assist in installation of IoT-based water management systems

To be competent, the user/individual on the job must be able to:

- PC12.** unpack, inspect and assemble IoT-enabled components as per guidelines
- PC13.** assist in setting up sensors and controllers for smart water systems
- PC14.** follow basic instructions to connect IoT devices with the main plumbing system
- PC15.** test the functionality of IoT devices with assistance
- PC16.** record and report any issues or irregularities in the IoT system

Knowledge and Understanding (KU)



Qualification Pack

The individual on the job needs to know and understand:

- KU1.** basics of plumbing systems, including water supply and drainage
- KU2.** characteristics and applications of pipes, fixtures and fittings
- KU3.** different types of joints and seals used in plumbing
- KU4.** functionality and components of IoT-enabled water management systems
- KU5.** use and maintenance of basic hand and power tools
- KU6.** common testing methods for plumbing systems
- KU7.** hazards and appropriate safety measures in plumbing work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret diagrams and instructions
- GS2.** communicate effectively with team members
- GS3.** organize tools and materials for efficient workflow
- GS4.** follow safety practices and use PPE
- GS5.** identify and escalate problems or irregularities



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for plumbing installations</i>	10	20	-	3
PC1. unpack and inspect pipes, fittings and fixtures as per safety and manufacturer guidelines	-	-	-	-
PC2. check the availability and specifications of materials, tools and equipment	-	-	-	-
PC3. identify missing or defective items and report them promptly	-	-	-	-
PC4. clean and organize the work area, ensuring tools and equipment are positioned for ease of access	-	-	-	-
<i>Assist in installation of plumbing systems</i>	10	20	-	4
PC5. use basic hand and power tools for cutting, bending and shaping pipes	-	-	-	-
PC6. assist in measuring and marking installation points	-	-	-	-
PC7. hold and position pipes and fixtures during installation	-	-	-	-
PC8. apply basic jointing techniques, such as threading, gluing and sealing under supervision	-	-	-	-
PC9. secure installed fixtures and pipes using clamps and fasteners	-	-	-	-
PC10. assist in testing plumbing systems for leaks and other issues as instructed	-	-	-	-
PC11. ensure proper insulation and alignment of pipes during installation	-	-	-	-
<i>Assist in installation of IoT-based water management systems</i>	10	20	-	3
PC12. unpack, inspect and assemble IoT-enabled components as per guidelines	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. assist in setting up sensors and controllers for smart water systems	-	-	-	-
PC14. follow basic instructions to connect IoT devices with the main plumbing system	-	-	-	-
PC15. test the functionality of IoT devices with assistance	-	-	-	-
PC16. record and report any issues or irregularities in the IoT system	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N1521
NOS Name	Assist in Installation of Plumbing Systems
Sector	Construction
Sub-Sector	
Occupation	MEP (Mechanical, Electrical and Plumbing)
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ICE/CON/N1522: Assist in Repair and Maintenance of Plumbing Systems

Description

This unit covers the competencies and knowledge requirements regarding the repair, maintenance, and sustainable management of plumbing systems.

Scope

The scope covers the following :

- Assist in repairing plumbing systems
- Assist in water quality and waste management
- Implement green plumbing practices

Elements and Performance Criteria

Assist in repairing plumbing systems

To be competent, the user/individual on the job must be able to:

- PC1.** shut off the water supply to the system under repair as instructed
- PC2.** dismantle pipes, fixtures or components safely under supervision
- PC3.** assist in identifying damaged or worn-out parts
- PC4.** assist in applying temporary fixes like sealing or patching leaks
- PC5.** replace damaged components using the correct tools and techniques
- PC6.** reassemble and secure repaired systems
- PC7.** test repaired systems for proper functioning

Assist in water quality test and waste management

To be competent, the user/individual on the job must be able to:

- PC8.** assist in installing basic filtration systems for water purification
- PC9.** record and report water quality issues as observed
- PC10.** assist in installing and maintaining greywater recycling systems
- PC11.** collect and segregate plumbing waste for proper disposal

Implement green plumbing practices

To be competent, the user/individual on the job must be able to:

- PC12.** assist in installing low-flow fixtures and water-saving devices
- PC13.** apply basic insulation to pipes to reduce energy usage
- PC14.** follow basic water conservation practices, e.g. repairing leaks to minimize water wastage

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** common causes of plumbing system failures



Qualification Pack

- KU2.** techniques for leak detection and repair
- KU3.** importance of sustainable waste management in plumbing
- KU4.** basic principles of water purification and filtration
- KU5.** green plumbing practices, including water conservation techniques
- KU6.** types of eco-friendly plumbing fixtures and their benefits
- KU7.** methods for maintaining plumbing tools and materials
- KU8.** importance of following environmental and safety regulations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate issues and solutions effectively
- GS2.** organize and prioritize tasks during repair work
- GS3.** interpret instructions and safety guidelines accurately
- GS4.** identify and report environmental hazards
- GS5.** apply problem-solving skills to minor plumbing challenges



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Assist in repairing plumbing systems</i>	10	20	-	3
PC1. shut off the water supply to the system under repair as instructed	-	-	-	-
PC2. dismantle pipes, fixtures or components safely under supervision	-	-	-	-
PC3. assist in identifying damaged or worn-out parts	-	-	-	-
PC4. assist in applying temporary fixes like sealing or patching leaks	-	-	-	-
PC5. replace damaged components using the correct tools and techniques	-	-	-	-
PC6. reassemble and secure repaired systems	-	-	-	-
PC7. test repaired systems for proper functioning	-	-	-	-
<i>Assist in water quality test and waste management</i>	10	20	-	4
PC8. assist in installing basic filtration systems for water purification	-	-	-	-
PC9. record and report water quality issues as observed	-	-	-	-
PC10. assist in installing and maintaining greywater recycling systems	-	-	-	-
PC11. collect and segregate plumbing waste for proper disposal	-	-	-	-
<i>Implement green plumbing practices</i>	10	20	-	3
PC12. assist in installing low-flow fixtures and water-saving devices	-	-	-	-
PC13. apply basic insulation to pipes to reduce energy usage	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. follow basic water conservation practices, e.g. repairing leaks to minimize water wastage	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ICE/CON/N1522
NOS Name	Assist in Repair and Maintenance of Plumbing Systems
Sector	Construction
Sub-Sector	
Occupation	MEP (Mechanical, Electrical and Plumbing)
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Awarding Body. Each Performance Criteria (PC)/ Element will be assigned marks proportional to its importance in NOS. AB will also lay down proportion of marks for Theory and Practical Skills for each PC/ Element.
2. The assessment of the knowledge part will be based on knowledge bank of questions created by Assessment Bodies subject to approval by AB.
3. Individual assessment agencies will create unique question papers for knowledge/theory part for assessment of candidates as per assessment criteria given below.
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on assessment criteria.
5. To pass the Qualification Pack, every trainee must score 70% on overall QP.
6. The Assessor shall check the outcome of the practices while evaluating the steps performed to achieve the outcome.
7. The trainee shall be provided with a chance to repeat the test to correct his procedures in case of



Qualification Pack

improper performance, with a deduction of marks for each iteration.

8. After the certain number of iterations as decided by AB the trainee is marked as fail, scoring zero marks for the procedure for the practical activity.

9. In case of successfully passing only certain number of NOS's, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack within the specified time frame set by AB.

10. Minimum duration of Assessment of each QP shall be 8hrs/batch (max. 30 candidates).

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ICE/CON/N1523. Identify Tools, Equipment and Materials Relevant to MEP Work at a Construction Site	30	60	0	10	100	15
ICE/CON/N1524. Read and interpret MEP system Drawings and Specifications	30	60	0	10	100	20
ICE/CON/N9901. Implement Safe Work Practices and Environmental Stewardship at Construction Sites	30	60	-	10	100	10
DGT/VSQ/N0101. Employability Skills (30 Hours)	20	30	-	-	50	5
Total	110	210	-	30	350	50

Elective: 1 Mechanical Systems



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ICE/CON/N1517.Assist in Installation of Mechanical Systems and HVAC Ducts	30	60	0	10	100	25
ICE/CON/N1518.Assist in Repair, Maintenance and Testing of Mechanical Systems	30	60	0	10	100	25
Total	60	120	-	20	200	50

Elective: 2 Electrical Systems

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ICE/CON/N1519.Assist in Electrical Wiring Low Voltage (LV) Installations and Earthing systems	30	60	0	10	100	25
ICE/CON/N1520.Assist in Installation, Repair and Maintenance of Electrical Systems	30	60	0	10	100	25
Total	60	120	-	20	200	50

Elective: 3 Plumbing Systems

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ICE/CON/N1521.Assist in Installation of Plumbing Systems	30	60	0	10	100	25
ICE/CON/N1522.Assist in Repair and Maintenance of Plumbing Systems	30	60	0	10	100	25



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	60	120	-	20	200	50



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.